Advanced Text Analytics

Supplementary Lectures

Master Degree Computer Engineering
Department of Information Engineering
University of Padua

November 5th, 2026

Introduction

Teacher

Marco Martinelli, MSc Computer Engineer, BSc Computer Engineer

3rd year PhD candidate, Department of Information Engineering, IIIA Hub Research Group

RESEARCH FIELDS

- Information Extraction
- Natural Language Processing
- Automated Knowledge Base Construction

CONTACTS

- Webpage: https://www.dei.unipd.it/~martinell2/
- Mail: marco.martinelli.4@phd.unipd.it
- Always sign your e-mails with your full name and prepend “[ATA2026]” in the subject

About these supplementary lectures

WHAT ARE WE GOING TO SEE

- Introduction to the use of Transformers and LLMs for Information Extraction

TASKS

- Token classification: Term Extraction (TE), Named Entity Recognition (NER)
- Definition Generation

RESOURCES

- Slides + scientific articles
- Python scripts + jupyter notebooks
- GitHub + websites

About these supplementary lectures

THESIS OPPORTUNITIES

- If you are interested in doing your master's thesis in NLP and working with LLMs, I have plenty of opportunities available.
- Wait for the end of these supplementary lectures and see if these things are of interest for you.

PARTICIPATION IN SHARED TASK(S)

- The datasets we work on are proposed as **shared tasks in international conferences**.
- This means you can try to get publications from working on these.

Options for Lectures

MOST PRACTICAL

- Will allow us to get hands-on (hopefully) with more than DETECH and GutBrainIE
- Drawback 1: Less theory ≈ Less understanding
- Drawback 2: Linked to D1, includes important pre-requisites:
    - Embeddings
    - Static vs Contextualized Embedding
    - Attention and Transformers
    - BERT
    - Language Models (and Large Language Models)

Options for Lectures

MORE BALANCED

- Will probably get us to just deal with DETECH and GutBrainIE
- A bit more theory compared to other option but still high-level and “code” oriented
- Drawback 1: More theory ≈ More boring (?)
- Drawback 2: We’ll see probably a bit less things, especially the ones with LLMs

Attention and Transformers

"The true art of memory is the art of attention"

(Samuel Johnson, Idler #74, 1759)

https://web.stanford.edu/~jurafsky/slp3/slides/transformer24aug.pdf

https://web.stanford.edu/~jurafsky/slp3/8.pdf

Introduction to Transformers

- Transformer is the standard architecture for building LLMs
- A transformer is a specific kind Neural Network (NN), a sort of feedforward NN but based on **attention**



Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com

Noam Shazeer* Google Brain noam@google.com

Niki Parmar* Google Research nikip@google.com

Jakob Uszkoreit* Google Research usz@google.com

Llion Jones* Google Research llion@google.com

Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu

Łukasz Kaiser* Google Brain lukaszkaiser@google.com

Illia Polosukhin* ‡ illia.polosukhin@gmail.com

Abstract

https://proceedings.neurips.cc/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

Attention is all you need

A Vaswani, N Shazeer, N Parmar… - Advances in neural …, 2017 - proceedings.neurips.cc

… to attend to **all** positions in the decoder up to and including that position. **We need** to prevent … **We** implement this inside of scaled dot-product **attention** by masking out (setting to −∞) …

☆ Save Cite Cited by 201960 Related articles All 70 versions »

From static to contextualized embeddings

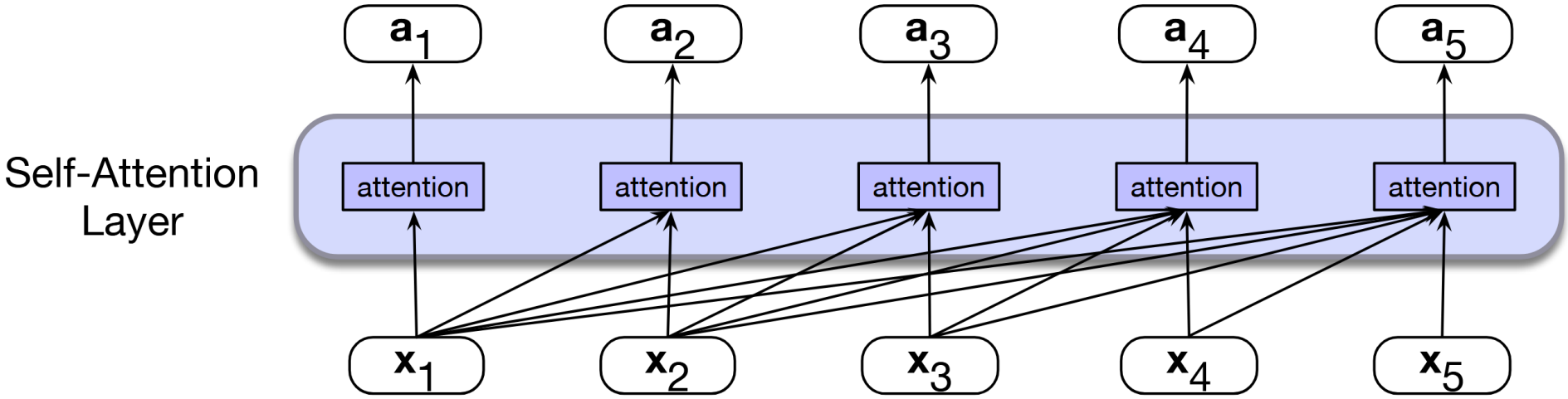
- Static embeddings (e.g., word2vec): a word is associated with a fixed embedding
- Problem: they are static!! The embedding for a word doesn't reflect how its meaning changes in context

  - I'm going to the **bank** to cash a check.
  - I'm going to the river **bank** for fishing.

- Intuition: the embedding for a word should represent its contextualized meaning
- Contextual embeddings (e.g., BERT): each word has a different vector that expresses different meanings depending on the surrounding words
- Attention is a method for computing contextual embeddings

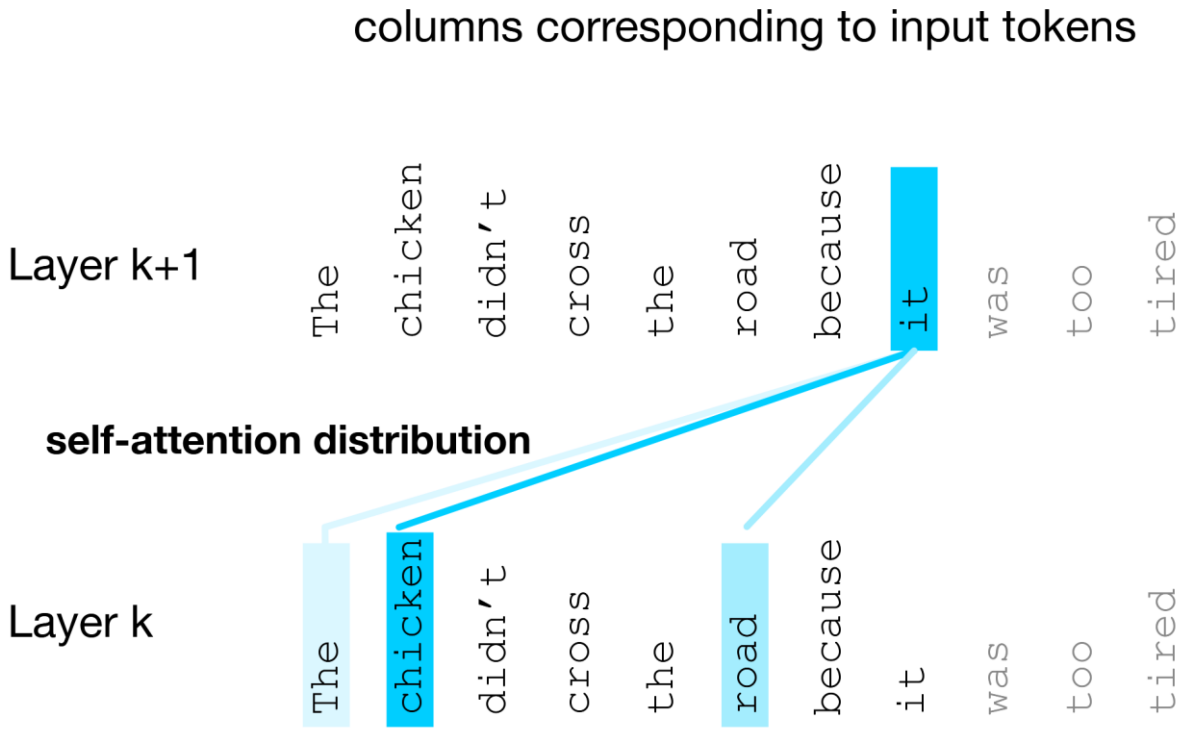
Intuition of attention

- Build the (contextual) embedding for a word by selectively integrating information from neighboring words. Not all neighboring words have the same importance.
- Attention is (natively) left to right

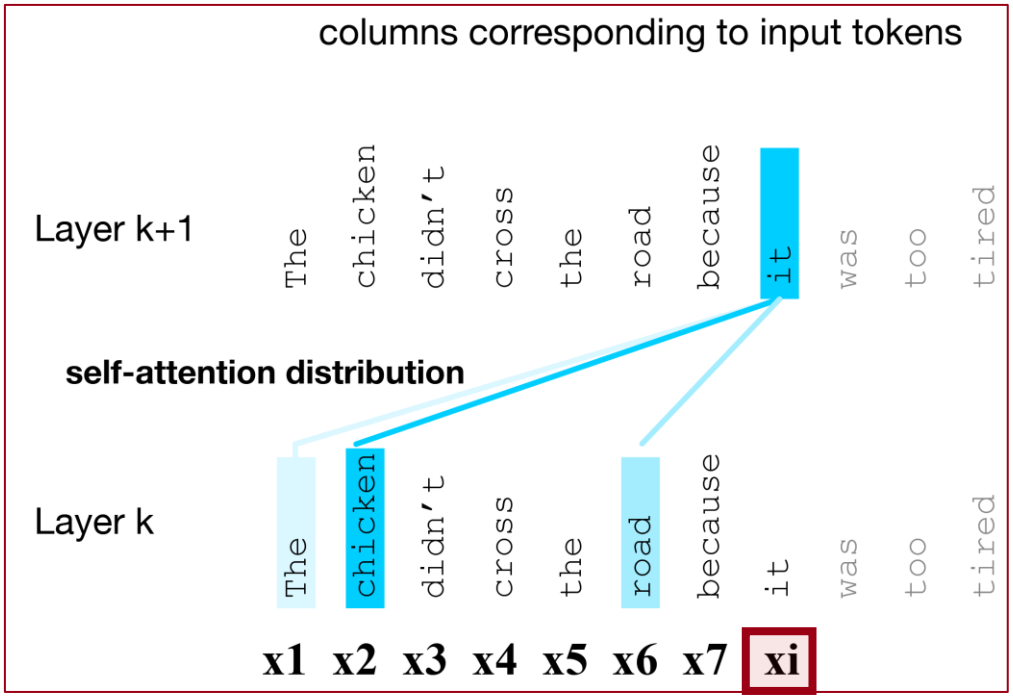


Example of attention

The chicken didn't cross the road because **it** was too tired



Attention Head



NAIVE IDEA

Vector x_i can be seen as a sum of prior words weighted by their similarity with the current word.

Start: sequence of token embeddings $[x_1, x_2, \dots, x_i]$

Score each token: $\forall j \leq i$

$$score(x_i, x_j) = x_i \cdot x_j$$

Normalize scores:

$$\alpha_{ij} = softmax\left(score(x_i, x_j)\right) \forall j \leq i$$

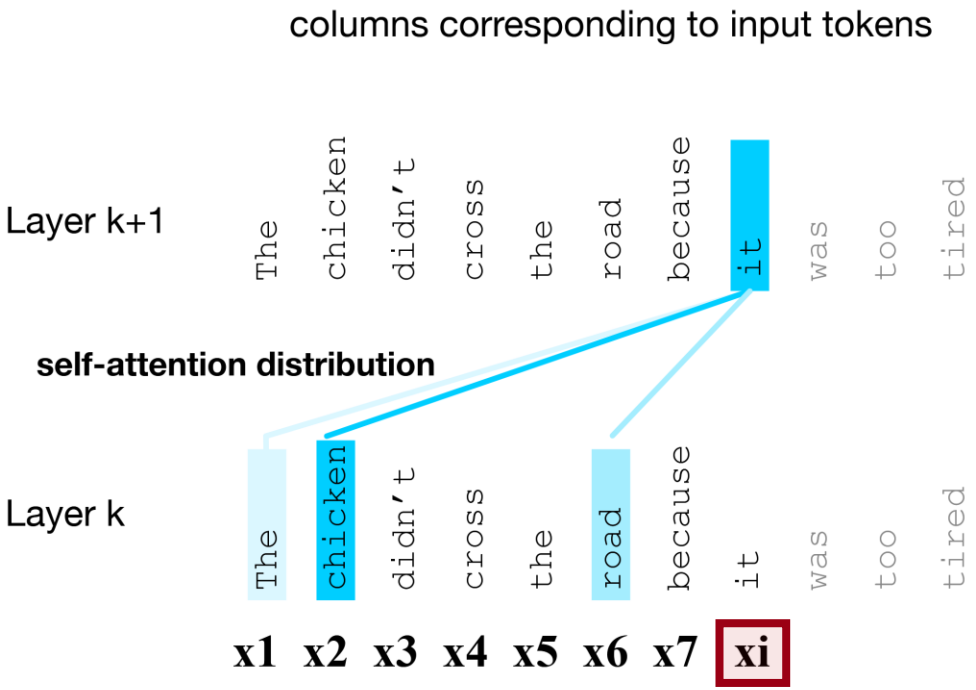
Compute attention for x_i:

$$a_i = \sum_{j \leq i} \alpha_{ij} \cdot x_j$$

Compute contextual embedding for x_i:

$$x_{i,contextual} = a_i \cdot x_i$$

Attention Head

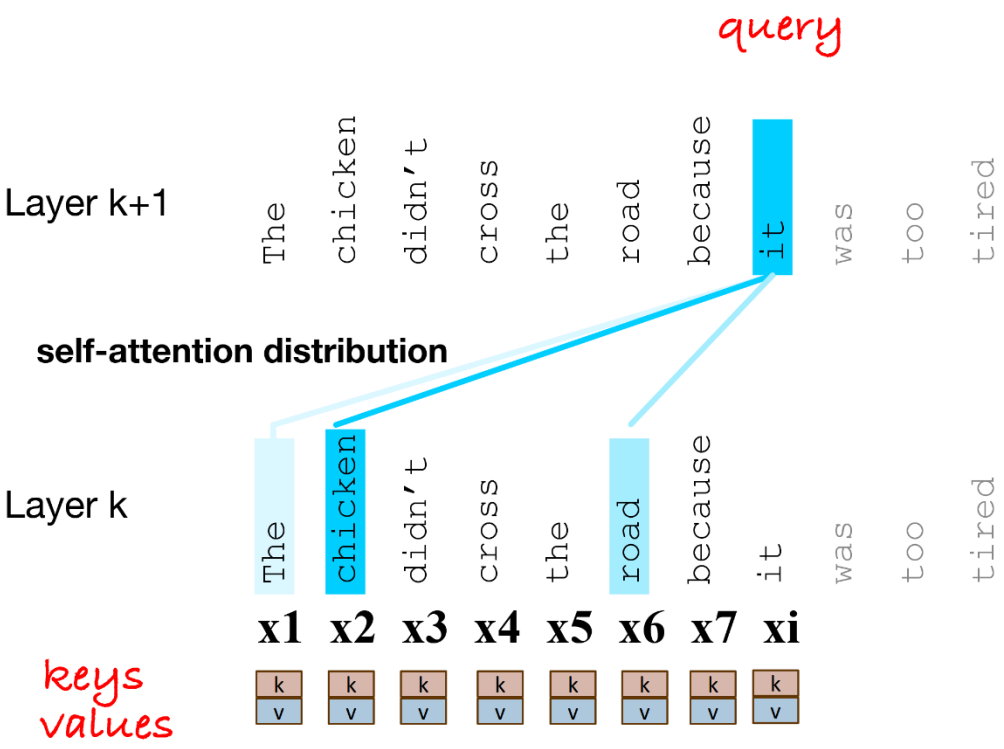


MORE FORMAL IDEA

Vector x_i can be seen as:

1. Query q: the current element being compared to the preceding inputs
2. Key k: a preceding input that is being compared to the current element to determine a similarity
3. Value v: a value of a preceding element that gets weighted and summed

Attention Head

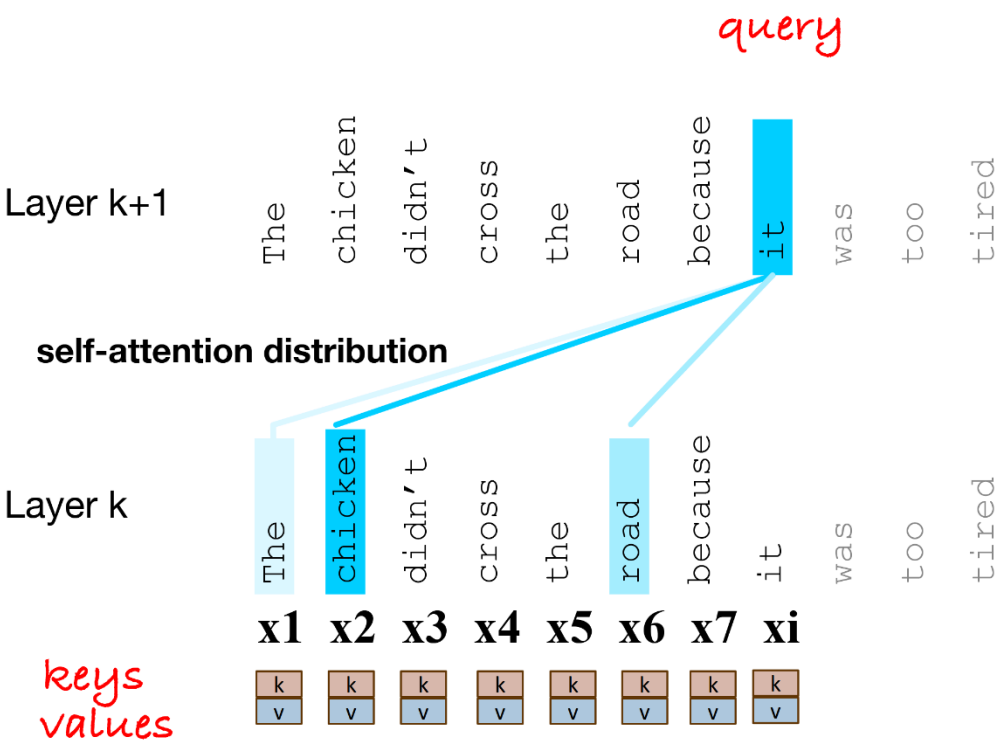


MORE FORMAL IDEA

Vector x_i can be seen as:

1. Query q: the current element being compared to the preceding inputs
2. Key k: a preceding input that is being compared to the current element to determine a similarity
3. Value v: a value of a preceding element that gets weighted and summed

Attention Head



MORE FORMAL IDEA

We can use matrices to project each vector x_i into a representation of its role as:

1. Query: W^Q
2. Key: W^K
3. Value: W^V

This gives 3 representation of x_i:

1. $q_i = x_i \cdot W^Q$
2. $k_i = x_i \cdot W^K$
3. $v_i = x_i \cdot W^V$

Attention Head

NAÏVE IDEA

Vector x_i can be seen as a sum of prior words weighted by their similarity with the current word:

Start: sequence of token embeddings $[x_1, x_2, \dots, x_i]$

Score each token: $\forall j \leq i$

$$score(x_i, x_j) = x_i \cdot x_j$$

Normalize scores:

$$\alpha_{ij} = softmax\left(score(x_i, x_j)\right) \forall j \leq i$$

Compute attention for x_i:

$$a_i = \sum_{j \leq i} \alpha_{ij} \cdot x_j$$

Compute contextual embedding for x_i:

$$x_{i,contextual} = a_i \cdot x_i$$

MORE FORMAL IDEA

3 representation of x_i:

$$1)\ q_i = x_i \cdot W^Q \quad 2)\ k_i = x_i \cdot W^K \quad 3)\ v_i = x_i \cdot W^V$$

Score each token: $\forall j \leq i$

$$score(x_i, x_j) = \frac{q_i \cdot k_j}{\sqrt{d_k}}$$

Normalize scores:

$$\alpha_{ij} = softmax\left(score(x_i, x_j)\right) \forall j \leq i$$

Compute attention for x_i:

$$a_i = \sum_{j \leq i} \alpha_{ij} \cdot v_j$$

Compute contextual embedding for x_i:

$$x_{i,contextual} = a_i \cdot x_i$$

Attention Multi-Head

- Instead of having a single attention head, we introduce a number h of attention heads

- Purpose: each head might capture different aspects of the context (e.g., gender, part of speech, co-reference…)

- For more precise and detailed information about attention and transformer, please refer to the resources reported in previous slides



BERT

> “What we call the beginning is often the end. And to make an end is to make a beginning. The end is where we start from.”
>
> (W. Sellard, 1991)

https://nlp.stanford.edu/seminar/details/jdevlin.pdf

https://harvard-iacs.github.io/CS287/lectures/11_BERT.pdf

https://cgi.csc.liv.ac.uk/~hang/ppt/BERT%20and%20Transformer.pdf

Introduction to BERT

- **B**idirectional **E**ncoder **R**epresentations from **T**ransformers



BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding

Jacob Devlin **Ming-Wei Chang** **Kenton Lee** **Kristina Toutanova**
Google AI Language
{jacobdevlin,mingweichang,kentonl,kristout}@google.com

https://aclanthology.org/N19-1423.pdf

Bert: Pre-training of deep bidirectional transformers for language understanding

J Devlin, MW Chang, K Lee… - Proceedings of the 2019 …, 2019 - aclanthology.org

We introduce a new language representation model called BERT, which stands for Bidirectional Encoder Representations from Transformers. Unlike recent language representation models (Peters et al., 2018a; Radford et al., 2018), BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers. As a result, the pre-trained BERT model can be fine-tuned with just one additional output layer to create state-of-the-art models for a wide range of …

☆ Save Cite Cited by 148398 Related articles All 54 versions »

Introduction to BERT



- **B**idirectional **E**ncoder **R**epresentations from **T**ransformers

- **B**idirectional: Like Bidirectional LSTMs, looks in **both** directions
- **E**ncoder … **T**ransformers: Uses only Transformer Encoders, no Decoders
- **R**epresentations: Builds rich representations

Problem with Previous Methods

- I'm going to the **bank** to cash a check.
- I'm going to the river **bank** for fishing.

Considering the examples above, a LM which uses only left context is not able to capture the difference between the two «bank» occurrences

Why are LMs unidirectional?

Directionality is needed to generate a well-formed probability distribution

In certain tasks we don't care about this and we are more interested in having words that can «see themselves»

Problem with Previous Methods

- I'm going to the **bank** to cash a check.
- I'm going to the **bank** of the river for fishing.

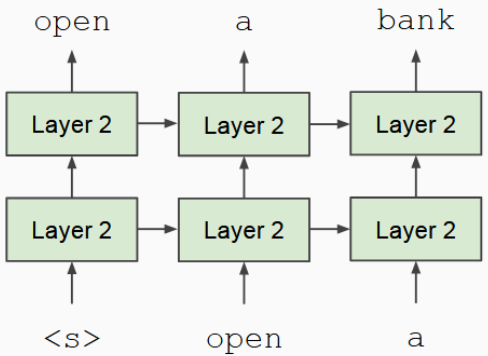
Cons…ples above, a LM which uses only left (or right) context is not able…

Why…

In ce…ords that…

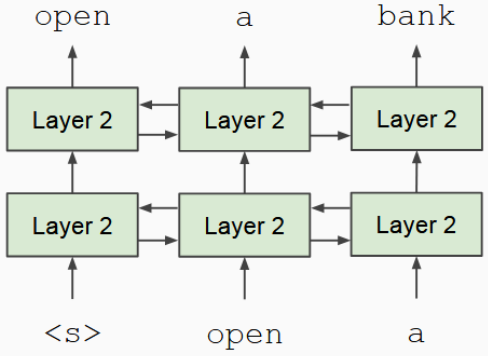
Unidirectional context

Build representation incrementally



Bidirectional context

Words can "see themselves"

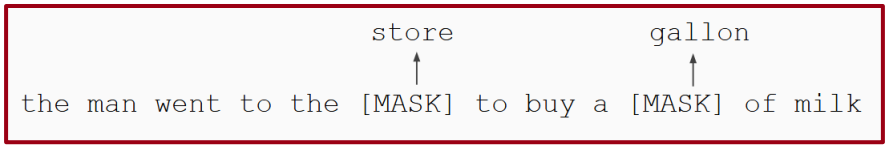


Introduction to BERT

BERT has 2 training objectives

1. Masked Language Modelling

- Goal: Predict a masked word
- Solution: mask out $k\%$ of the input words, then predict the masked words
- $k = 15\%$; 80% of replacements with [MASK] ; 10% of replacements with random word ; 10% with original word (i.e., keep the same word)



2. Next Sentence Prediction

- Goal: Given two subsequent sentences A and B, predict if sentence B truly follows sentence A or not.
- Solution: concatenate the two sentences and train the model to predict one of two labels «isNext» or «notNext»
- Alternative view: predict the likelihood that sentence B belongs after sentence A

Introduction to BERT

BERT has 2 training objectives

1. Masked Language Modelling

- Goal: Predict a masked word
- Solution: mask out $k\%$ of the input words, then predict the masked words
- $k = 15\%$; 80% of replacements with [MASK] ; 10% of replacements with random word ; 10% with original word (i.e., keep the same word)



2. Next Sentence Prediction

- Goal: Given two subsequent sentences, predict if the second sentence truly follows the first one or not.
- Solution: concatenate the two sentences

Sentence A = The man went to the store.
Sentence B = He bought a gallon of milk.
Label = IsNextSentence

Sentence A = The man went to the store.
Sentence B = Penguins are flightless.
Label = NotNextSentence

BERT for Token Classification

> “Any displayed token has countless features, and so it is of countless different kinds.”
>
> (Bennett, 1988)

https://huggingface.co/tasks/token-classification

https://ceur-ws.org/Vol-4038/paper_5.pdf

https://huggingface.co/docs/transformers/tasks/token_classification

https://www.maxwell.vrac.puc-rio.br/16458/16458_4.PDF

BERT for Token Classification

WHAT IS TOKEN CLASSIFICATION

- It is a common NLP task that assigns a label or class to some tokens in a text.
- Formally, the task is formulated as labeling each token with a class, using class “O” for tokens that “are not of interest” (Outside)
- Two main subtasks: Term Extraction (TE), Named Entity Recognition (NER)

HOW CAN TOKEN CLASSIFICATION BE APPROACHED WITH BERT

- Get a representation of tokens
- Attach a classification head on top (linear with output layer size = # labels) to classify each token

BERT for Token Classification

APPROACH

We can model token classification using the Beginning-Inside-Outside (BIO) tagging schema.

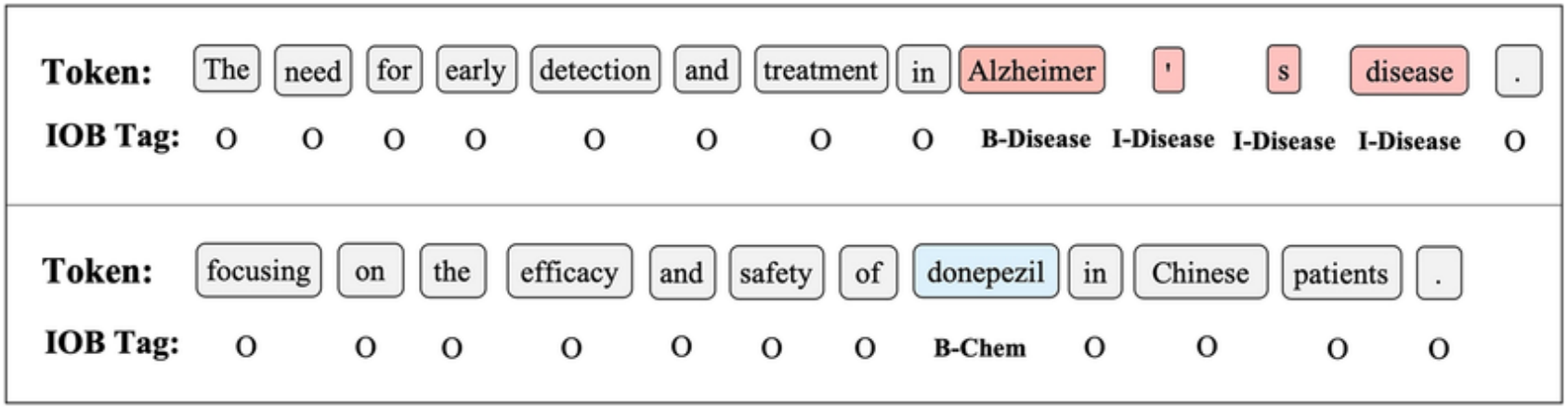
In this schema, tokens are labeled with one of three prefixes:

- "**B-***" indicating the **Beginning** of an entity
- "**I-***" indicating a token **Inside** an entity
- "**O**" indicating a token that does not belong (**Outside**) to any entity

BERT for Token Classification

APPROACH

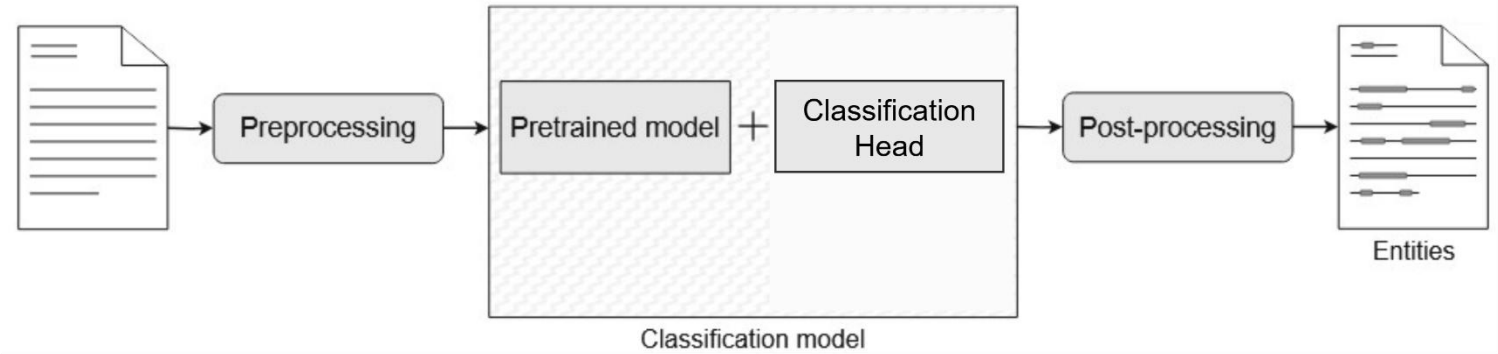
We can model token classification using the Beginning-Inside-Outside (BIO) tagging schema.



Alamro, Hind & Gojobori, Takashi & Essack, Magbubah & Gao, Xin. (2024). BioBBC: a multi-feature model that enhances the detection of biomedical entities. Scientific Reports. 14. 10.1038/s41598-024-58334-x.

BERT for Token Classification

PIPELINE



Andersen, Lasse Ryge, et al. "Trusting Gut Instincts: Transformer-Based Extraction of Structured Data from Gut-Brain Axis Publications." CLEF (2025).

BERT for Token Classification

EVALUATION METRICS

	Correct	Not Correct
Selected	True Positives	False Positives
Not Selected	False Negatives	True Negatives

Let TP_ℓ , FP_ℓ , and FN_ℓ denote, respectively, the number of True Positives, False Positives, and False Negatives for label ℓ. We define the label set $\mathcal{L}$ as:

- For TE: a single label set
- For NER: the set of entity types

BERT for Token Classification

EVALUATION METRICS

	Correct	Not Correct
Selected	True Positives	False Positives
Not Selected	False Negatives	True Negatives

- Precision: $P = \frac{TP}{TP+FP} = \frac{True\ Positives}{True\ Positives + False\ Positives}$
- Recall: $R = \frac{TP}{TP+FN} = \frac{True\ Positives}{True\ Positives + False\ Negatives}$
- F1-score: $F_1 = \frac{2 \times P \times R}{P+R} = \frac{2 \times Precision \times Recall}{Precision + Recall}$

BERT for Token Classification

EVALUATION METRICS

	Correct	Not Correct
Selected	True Positives	False Positives
Not Selected	False Negatives	True Negatives

$P_{macro} = \frac{1}{\lvert\mathcal{L}\rvert}\sum_{\ell\in\mathcal{L}}\frac{TP_\ell}{TP_\ell + FP_\ell}$	$P_{micro} = \frac{\sum_{\ell\in\mathcal{L}} TP_\ell}{\sum_{\ell\in\mathcal{L}}(TP_\ell + FP_\ell)}$
$R_{macro} = \frac{1}{\lvert\mathcal{L}\rvert}\sum_{\ell\in\mathcal{L}}\frac{TP_\ell}{TP_\ell + FN_\ell}$	$R_{micro} = \frac{\sum_{\ell\in\mathcal{L}} TP_\ell}{\sum_{\ell\in\mathcal{L}}(TP_\ell + FN_\ell)}$
$F1_{macro} = \frac{1}{\lvert\mathcal{L}\rvert}\sum_{\ell\in\mathcal{L}}\frac{2P_\ell R_\ell}{P_\ell + R_\ell}$	$F1_{micro} = \frac{2P_{micro}R_{micro}}{P_{micro} + R_{micro}}$